

Name: Answer

Exam Score: _____

1. (16 pt) For the following parts, $A = \begin{bmatrix} 1 & 0 & -3 & 0 & 2 \\ -2 & 1 & 7 & 0 & -1 \\ 0 & -2 & -2 & 1 & -7 \\ 3 & 1 & -8 & 0 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 1 & 7 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = B$

$v_1 \quad v_2 \quad v_3 \quad v_4 \quad v_5$

a) Fill in the following blanks:

(6) (i) $\text{rank}(A) = \underline{3}$ (ii) $\dim(\text{Nul}(A)) = \underline{2}$ (iii) $\dim(\text{Nul}(A^T)) = \underline{1}$

b) Give a basis for the null space of A .

(7) $B \sim \begin{bmatrix} 1 & 0 & -3 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{aligned} x_1 &= 3x_3 - 2x_5 \\ x_2 &= -x_3 - 3x_5 \\ x_4 &= x_5 \end{aligned}$

$\Rightarrow x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3x_3 - 2x_5 \\ -x_3 - 3x_5 \\ x_3 \\ x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 3 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad x_3, x_5 \in \mathbb{R}$

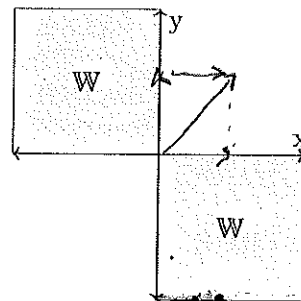
Basis for $\text{Nul}(A)$ is $\begin{bmatrix} 3 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

c) Give a basis for the column space of A .

(3) $v_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 1 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

2. (8 pt) Circle TRUE or FALSE for each of the following statements concerning the set $W = \{(x, y) \in \mathbb{R}^2 \mid xy \leq 0\}$:

- a) TRUE FALSE $u + v \in W$ for every $u, v \in W$.
- b) TRUE FALSE $\alpha v \in W$ for every $\alpha \in \mathbb{R}$ and every $v \in W$.
- c) TRUE FALSE $0 = (0, 0) \in W$.
- d) TRUE FALSE W is a subspace of the vector space $\mathbb{R}^2$.



3. (4 pt) V is a vector space and B is a set of vectors in V . Give the two conditions that guarantee that B is a basis for V .

(i) B is an independent set

(ii) $V = \text{span}(B)$

4. (10 pt) $B = \{p_1(t) = t, p_2(t) = 2t + 1, p_3(t) = t^2 - t + 1\}$ is a basis for $\mathbb{P}_2$. Find the coordinates of $p(t) = 2t^2 - t + 1$ with respect to the basis B .

$$P_{\mathcal{B}} [p]_{\mathcal{B}} = [p]_{\mathcal{B}} \Rightarrow \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix} [p]_{\mathcal{B}} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 0 & 1 & 1 & 1 \\ 1 & 2 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} \mathbf{I}_3 & & & \begin{matrix} 3 \\ -1 \\ 2 \end{matrix} \end{array} \right] \Rightarrow [p]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$$

$$p(t) = 3t - (2t+1) + 2(t^2 - t + 1)$$

5. (12 pt) For the following statements circle **TRUE** if the statement is always true; otherwise circle **FALSE**:

- a) **TRUE** FALSE If A is an $m \times n$ matrix and E is an invertible $m \times m$ matrix, then $\text{Row}(A) = \text{Row}(EA)$.
- b) **TRUE** FALSE An $n \times n$ matrix A is invertible if and only if 0 is not an eigenvalue of A .
- c) **TRUE** FALSE Similar $n \times n$ matrices have the same eigenvalues.
- d) **TRUE** **FALSE** An $n \times n$ matrix A is diagonalizable if and only if it has n different eigenvalues.
- e) **TRUE** FALSE V is a vector space and $\dim(V) = n$. If $\mathcal{S} = \{v_1, \dots, v_n\} \subset V$ is linearly independent, then $\mathcal{S}$ is a basis for V .
- f) **TRUE** **FALSE** If $T: V \rightarrow W$ is a linear transformation between two vector spaces V and W , then $\mathcal{S}$ is a basis for V if and only if $T(\mathcal{S})$ is a basis for W .

6. (12 pt) Let $A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. Then $\det(\lambda I_3 - A) = (\lambda - 1)^2(\lambda - 2)$, and $\text{Nul}(A - 2I_3) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$.

If A can be diagonalized, give a diagonal matrix D and an invertible matrix P such that $A = PDP^{-1}$. If A cannot be diagonalized, then explain why not.

$$A - 2I = \begin{bmatrix} -2 & 1 & -1 \\ -1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow x_1 - x_2 + x_3 = 0$$

$$\Rightarrow x_1 = x_2 - x_3 \Rightarrow x = \begin{bmatrix} x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{So with } P = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Then } A = PDP^{-1}$$

7. (12 pt) $A = \begin{bmatrix} 4 & -5 \\ 1 & 0 \end{bmatrix}$. Note that $\lambda = 2 - i$ is an eigenvalue of A with eigenvector $v = \begin{bmatrix} 2 - i \\ 1 \end{bmatrix}$.

a) Give a diagonal matrix $D \in \mathbb{C}^{2 \times 2}$ and invertible matrix $P \in \mathbb{C}^{2 \times 2}$ such that $A = PDP^{-1}$.

$$\textcircled{6} \quad A - (2-i)I = \begin{bmatrix} 2+i & -5 \\ 1 & -2+i \end{bmatrix} \sim \begin{bmatrix} 1 & -2+i \\ 0 & 0 \end{bmatrix} \Rightarrow x_1 = (2-i)x_2 \Rightarrow x = x_2 \begin{bmatrix} 2-i \\ 1 \end{bmatrix} \checkmark$$

$$A = PDP^{-1} \text{ when } P = \begin{bmatrix} 2-i & 2+i \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 2-i & 0 \\ 0 & 2+i \end{bmatrix}$$

b) Give a matrix $Q = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ and an invertible matrix $P \in \mathbb{R}^{2 \times 2}$ such that $A = PQP^{-1}$.

$$\textcircled{6} \quad A = PQP^{-1} \text{ when } P = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}, Q = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

8. (26 pt) For the following parts, let $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$, $\mathcal{E} = \{e_1, e_2\}$ represent bases for $\mathbb{R}^2$. Where

$$b_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, c_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, c_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

a) Find the change-of-coordinate matrices given by ${}_{\mathcal{E} \leftarrow B}^P$ and ${}_{C \leftarrow \mathcal{E}}^P$.

$${}_{C \leftarrow \mathcal{E}}^P = {}_{\mathcal{E} \leftarrow C}^P = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}^{-1}$$

$$(2) {}_{\mathcal{E} \leftarrow B}^P = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

$$(4) {}_{C \leftarrow \mathcal{E}}^P = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$$

b) Find ${}_{C \leftarrow B}^P$, $[v]_{\mathcal{E}}$, and $[v]_C$, when $v = -2b_1 + b_2$.

$${}_{C \leftarrow B}^P = {}_{C \leftarrow \mathcal{E}}^P {}_{\mathcal{E} \leftarrow B}^P = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -3 & -5 \\ 5 & 8 \end{bmatrix}$$

$$-2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$(3) {}_{C \leftarrow B}^P = \begin{bmatrix} -3 & -5 \\ 5 & 8 \end{bmatrix}$$

$$(2) [v]_{\mathcal{E}} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$(3) [v]_C = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

c) Let $T(x) = Ax$ where $A = \begin{bmatrix} -2 & 1 \\ -9 & 4 \end{bmatrix}$. Find the B -matrix of T .

$$(6) [T]_B = {}_{B \leftarrow \mathcal{E}}^P A {}_{\mathcal{E} \leftarrow B}^P = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ -9 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

d) $M[v]_B = [Av]_C$ where $A = \begin{bmatrix} -2 & 1 \\ -9 & 4 \end{bmatrix}$. Find $M \in \mathbb{R}^{2 \times 2}$.

$$(6) M = {}_{C \leftarrow \mathcal{E}}^P A {}_{\mathcal{E} \leftarrow B}^P = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ -9 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -5 \\ -19 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -3 & -8 \\ 5 & 13 \end{bmatrix}$$